PASSKEY-FIRST SECURITY

Secure what happens after the password disappears.

Identity, recovery and team secrets protected by a zero-knowledge vault.

SN

IDENTITY
RECOVERY
TEAM SECRETS

Professional pitch | 15 July 2026

WHY NOW

Passkeys are mainstream. Security continuity is not.

5B

passkeys estimated in active use worldwide

FIDO Alliance, State of Passkeys 2026

75%

of surveyed consumers enabled a passkey on at least one account

FIDO Alliance, State of Passkeys 2026

68%

of surveyed organizations are deploying, piloting or rolling out workforce passkeys

FIDO Alliance, State of Passkeys 2026

The market is solving sign-in. The next category is managing recovery, devices and shared secrets without rebuilding a password-shaped product.

THE PROBLEM

Passwordless sign-in does not solve access continuity.

01

Identity fragments

Passkeys prove who is signing in, but recovery and device replacement remain separate, inconsistent workflows.

02

Secrets remain scattered

API keys, break-glass records and shared credentials still live in chat, documents and legacy vaults.

03

Teams lose context

Onboarding, offboarding and access changes are hard to audit when identity and secret governance are disconnected.

Credential abuse remains an entry path in 22% of breaches reviewed in Verizon's 2025 DBIR research.

Source: Verizon Business, Additional 2025 DBIR research on credential stuffing.

THE PRODUCT

One secure lifecycle: sign in, unlock, recover, share, govern.

01	02	03	04	05
Sign in	**Unlock**	**Recover**	**Share**	**Govern**
Passkey-first authentication	PRF-derived cryptographic access	User-held recovery kit	Encrypted team workspaces	Devices, roles and audit

SafeNode is the continuity and secret-governance layer for a passkey-first world.

CRYPTOGRAPHIC DIFFERENTIATION

The passkey is part of the key path - not a screen lock.

1 **WebAuthn assertion**

Authenticator verifies the user and evaluates the PRF extension.

2 **HKDF-SHA-256**

PRF output derives a credential-specific AES-256 key-encryption key.

3 **Vault key unwrap**

The KEK unwraps the random vault key in client memory.

4 **Local decryption**

AES-256-GCM decrypts vault ciphertext for the active session.

ZERO-KNOWLEDGE BOUNDARY

The server stores ciphertext, public credential data, salts and wrapped keys - never the vault key, recovery kit or PRF output.

PRODUCT SYSTEM

A security workspace, not a collection of settings.

Identity Vault

Personal records, passkey readiness and encrypted access state.

Recovery Center

Recovery kit, fallback readiness and successor continuity.

Team Secrets

Shared encrypted workspaces, roles, invitations and audit.

Trusted Devices

Registration, sessions, approval, revocation and device swaps.

Security Posture

Breach signals, weak credentials, risky devices and recovery gaps.

Reports and Audit

Operational history and exportable evidence for access changes.

WHY SAFENODE

Designed around key ownership and honest fallback.

01

Zero knowledge

Backend infrastructure cannot decrypt vault plaintext from stored data alone.

02

Passkey PRF

Authentication output derives the key used for cryptographic vault unlock.

03

Recovery without escrow

User-held recovery material restores the same random vault key locally.

04

No local master-key cache

Raw unlock secrets remain memory-only and transient fields never enter encrypted exports.

05

Migration safety

Legacy users upgrade without destructive reinitialization or forced PRF dependency.

06

Evidence-first releases

Tests, signer checks, domain association and download gates are automated.

BUSINESS MODEL

Land with personal security. Expand through continuity and teams.

FREE	PERSONAL	FAMILY	TEAMS
$0	**$2.49/mo**	**$4.16/mo**	**$8.32/mo**
One device, encrypted vault and breach monitoring.	Five devices, sync, passkeys and priority support.	Ten devices, shared vaults and family controls.	Admin workspace, RBAC, SSO and business support.

Current product configuration as of 15 July 2026. Pricing and packaging remain subject to launch validation.

GO TO MARKET

Start where access changes frequently and failure is expensive.

BEACHHEAD

1. Security-conscious operators

Founders, developers and consultants already using passkeys but still managing recovery and client secrets manually.

EXPAND

2. Remote SMB teams

Organizations without a dedicated IAM program that need clear onboarding, offboarding and shared access governance.

SCALE

3. Managed environments

Agencies and service providers managing multiple customer contexts, break-glass access and credential rotation.

Distribution loops

Developer communities | security education | team invitations | recovery-readiness assessments | partner integrations

EXECUTION EVIDENCE

The security thesis is implemented in code - with clear release boundaries.

60/60

frontend tests passing across 11 files

Verified locally on security/p0-hardening

151/151

backend tests passing across 23 suites

Verified against a dedicated PostgreSQL test database

4096-bit

stable Android release signing key

Signed APK verified with Android apksigner

HONEST RELEASE STATUS

Production hosting is currently blocked by Vercel fair-use enforcement. Android download remains intentionally disabled until API health, live Digital Asset Links and a newly signed artifact pass the device checklist.

ROADMAP

From strong foundation to enterprise-grade secret continuity.

NOW	**Restore production**	Resolve hosting, rotate exposed secrets, deploy API and web, publish live DAL and signed APK.
NEXT	**Multi-device assurance**	Validate PRF across real authenticators, synced passkeys and cross-browser fallback.
THEN	**Team wrapped keys**	Per-member key wrapping, targeted revocation, rotation and stronger cryptographic ACLs.
SCALE	**Enterprise controls**	Directory lifecycle, policy, richer audit evidence and independent security assurance.

THE PASSKEY ERA NEEDS A SECURITY LAYER

SafeNode protects identity continuity - before, during and after sign-in.

Passkey-first authentication. Zero-knowledge vaults. Recovery without escrow. Team secrets with accountable access.

safe-node.app
Architecture and partnership conversations

Sources: FIDO Alliance 2026; Verizon 2025 DBIR research; IBM 2025.

SOURCES AND CLAIM BOUNDARIES

Research used in this presentation.

FIDO ALLIANCE

The State of Passkeys 2026

https://fidoalliance.org/the-state-of-passkeys-2026-global-consumer-and-workforce-report/

FIDO ALLIANCE

Passkey Index 2025

https://fidoalliance.org/passkey-index-2025/

VERIZON BUSINESS

Additional 2025 DBIR research on credential stuffing

https://www.verizon.com/business/resources/articles/credential-stuffing-attacks-2025-dbir-research/

IBM

Cost of a Data Breach Report 2025

https://www.ibm.com/reports/data-breach

No customer count, revenue, growth, certification, penetration-test or compliance claims are made. Product readiness statements distinguish verified implementation from production blockers and roadmap.